

Distinguishing Features of Patients Evaluated for Multisystem Inflammatory Syndrome in Children

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Objectives: Given the significant overlap of multisystem inflammatory syndrome in children (MIS-C) with other common childhood illnesses presenting to the emergency department, extensive workup of this syndrome has become necessary. Nevertheless, little has been published on the factors differentiating MIS-C from other conditions in the acute care setting. We investigated differences in presentation and laboratory studies between suspected versus confirmed MIS-C patients.

Methods: This was a retrospective cohort study on patients 21 years or younger undergoing investigation for possible MIS-C at a single institution between April 21 and July 1, 2020. The primary outcome was diagnosis of MIS-C or an alternative final diagnosis. Clinical features and laboratory findings from initial presentation were collected and analyzed.

Results: A total of 106 patients (median, 4 years; 55.7% male) were included, of whom 17 (16%) of 106 met the criteria for MIS-C. Multisystem inflammatory syndrome in children patients were significantly more likely to report a coronavirus disease 2019 exposure (odds ratio (OR), 13.17 [3.87–44.9]), have gastrointestinal symptoms (OR, 3.81 [1.02–14.19]), and have a significantly higher odds of having abnormal laboratory values including high-sensitivity troponin T (OR, 13 [4.0–42.2]), N-terminal B-type natriuretic peptide (OR, 8.4 [2.3–30.1]), D-dimer (OR, 13 [1.6–103]), and ferritin (OR, 7.8 [2.2–27.2]). There were also differences between groups in inflammatory markers: C-reactive protein (median, 134.45 mg/L vs 12.6 mg/L; $P < 0.05$) and procalcitonin (1.71 ng/mL vs 0.14 ng/mL; $P < 0.001$).

Conclusions: Higher elevations in key laboratory studies may help to distinguish between MIS-C patients and non-MIS-C patients presenting to the emergency department.

Key Words: SARS-CoV-2, COVID-19, MIS-C, PIMS-TS, diagnostic criteria

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Initial analysis of the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) pandemic suggested that children were less susceptible with less severe presentations.¹ However, reports from Europe^{2–7} and the United States^{8,9} revealed previously healthy children presenting with a severe “Kawasaki-like” inflammatory syndrome with some children requiring intensive care, mechanical ventilation, and extracorporeal membrane oxygenation.^{3,4,9,10} Now identified as multisystem inflammatory syndrome in children (MIS-C), the Centers for Disease and Prevention Control (CDC)¹¹ and World Health Organization¹² have issued criteria for the diagnosis. These diagnostic criteria vary slightly but both hinge on 5 components: (1) fever, (2) multiorgan dysfunction, (3) laboratory evidence of inflammation, (4) exposure to and/or positive testing for

SARS-CoV-2, and (5) lack of clear alternative diagnoses. However, aside from exposure to and/or positive testing for SARS-CoV-2, the presenting symptoms and laboratory studies may vary widely.^{7–9,13} Furthermore, the turnaround time for SARS-CoV-2 polymerase chain reaction (PCR) and antibody testing differs among facilities, and because patients with either test may meet MIS-C criteria, both tests are usually indicated. Although exposure to family members who are SARS-CoV-2 positive is a clear risk factor, the prevalence of SARS-CoV-2 infection in some communities may be high enough to reasonably suspect exposure in any patient,¹⁴ limiting exposure history as a useful discriminatory evaluation tool.

Multisystem inflammatory syndrome in children encompasses a large clinical spectrum with likely recognizable severe disease features including cardiogenic shock, but the diagnosis of more mild disease presentations has thus far been less defined. Common childhood illnesses including streptococcal pharyngitis¹⁵ and viral gastroenteritis¹⁶ may present with symptoms similar to MIS-C, and before the SARS-CoV-2 era, extensive laboratory evaluation of children presenting with common viral illness was not routinely obtained. In these cases, using laboratory findings to discriminate possible MIS-C may not provide diagnostic clarity and may even mislead providers in severe cardiac outcomes such as coronary artery dilation.⁷

As the coronavirus disease 2019 (COVID-19) pandemic continues during the increasing seasonal prevalence of alternative pathogens, the diagnosis of MIS-C is critical to make among other presenting illnesses. Although there has been one report highlighting the differences between MIS-C and other febrile illnesses in the outpatient setting, no research has investigated this differentiation among initial presentations in higher acuity settings.¹⁷ Outpatient samples skew toward less severe presentations, raising questions about the generalizability to the pediatric emergency room setting. Emergency medicine providers must balance an accurate diagnosis of MIS-C against unnecessary hospitalizations and additional testing. To this end, we reviewed more than 100 pediatric patients under investigation (PUIs) for MIS-C presenting to care at our institution to determine the association among presenting symptoms, laboratory values and diagnosis of MIS-C.

METHODS

Study Population

We conducted a retrospective chart review to identify patients undergoing workup for possible MIS-C at quaternary care academic hospital between April 21, 2020, and July 1, 2020. April 21, 2020, was used as an initial date as this represented the first case of MIS-C at our institution,¹⁸ also correlating with the initial reports out of the United Kingdom.¹⁹

Criteria for identification of cases were patients 21 years or younger who had the following performed during the encounter: SARS-CoV-2 PCR, C-reactive protein (CRP), and high-sensitivity troponin T (hsTropT). Medical records that met these criteria were then manually reviewed to verify clinical concern for MIS-C (Fig. S1). This medical records research study was determined to be exempt by our institutional research board.

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Data Collection

All included medical records were manually reviewed for demographic information, clinical characteristics, laboratory values, clinical course, and final diagnosis. Temperature was recorded as the maximum temperature within the first 24 hours of presentation; all other vital signs were recorded as documented on initial presentation. The laboratory investigations analyzed were based on the case definitions for MIS-C.^{11,12} Presenting signs and symptoms were divided into the following categories: cardiovascular (chest pain), constitutional (chills, fatigue, fever, rigors), dermatologic (edema of hands or feet, erythema, rash), gastrointestinal (abdominal pain, loss of appetite, nausea, vomiting), mucocutaneous (conjunctivitis, mucosal changes), musculoskeletal (arthralgias, myalgias), neurologic (altered mental status, headache), and respiratory (congestion, cough, increased work of breathing, loss of taste or smell, shortness of breath, sore throat, rhinorrhea, wheezing).

Final diagnosis was determined via chart review of clinical documentation. The data were entered into the Research Electronic Data Capture, an electronic database hosted by our hospital.^{20,21}

Study Definitions

Normal values for heart rate, blood pressure, and obesity were defined based on age-based norms.²² Further details can be found in the supplemental appendix. The diagnosis of MIS-C was made in patients who met either the CDC or World Health Organization case definitions.^{11,12}

Statistical Analysis

Data were analyzed using IBM SPSS Statistics for Windows version 24.0 (IBM Corp, Armonk, NY) and SAS OnDemand for Academics 3.8 software (SAS Institute Inc, Cary, NC). Statistical significance for variables was defined as *P* value of <0.05. Categorical variables were described using frequencies and percentages; continuous variables were described using mean, median, standard deviation, and interquartile range. Categorical data were analyzed by χ^2 or Fisher exact tests accordingly. Parametric and nonparametric continuous data were analyzed by *t* tests and Mann-Whitney *U* tests, respectively. Significant laboratory values that were identified on univariate analysis (*P* < 0.05) were assessed further for odds ratio and the area under the curve. The laboratory values were converted to dichotomous variables using our laboratory norms. With regards to area under the curve, if the value was greater than 0.7, optimal cutoffs were chosen based on the Youden index^{23–25} through receiver operating characteristic (ROC) curves.

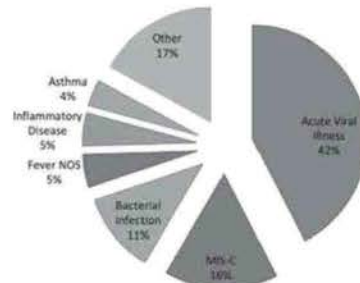


FIGURE 1. Final diagnosis of the population under investigation for MIS-C. Final diagnoses of the 106 PUIs for possible MIS-C spanned a wide range of diagnoses. The most common diagnoses were viral illnesses (45/106, 42%), including acute SARS-CoV-2 (6/106, 6%). Multisystem inflammatory syndrome in children was diagnosed in 17 (16%) of 106 patients. Bacterial infections were seen in 12 (11%) of 106 patients. All final diagnoses were made by chart review.

RESULTS

Patient Demographics and Presentation

One hundred six patients met the inclusion criteria, of whom 17 (16%) met the criteria for MIS-C. Forty-five patients (42%) were diagnosed with acute viral illnesses, and 12 patients (11%) were diagnosed with bacterial infections (Fig. 1). Baseline demographics for patients are shown in Table 1; no statistically significant differences in age, sex, weight, or obesity were seen between the MIS-C and patients with other diagnoses (non-MIS-C) cohorts.

Multisystem inflammatory syndrome in children patients were more likely to be previously healthy (OR [95% confidence interval], 3.18 [0.96–10.5]), be hospitalized (odds ratio [OR], 9.89 [1.25–78]), and require intensive care unit level of care (OR, 5.14 [1.57–16.88]). Among the presenting signs and symptoms, MIS-C patients were more likely to report COVID-19 exposure (OR, 13.17 [3.87–44.9]), present with at least one constitutional symptom (100% vs 66.3%; *P* < 0.01), and present with at least 1 gastrointestinal complaint (OR, 3.81 [1.02–14.19]) (Table 1; Table 2). Among vital signs, fever in MIS-C patients was borderline significant (OR, 2.81 [0.97–8.12]; *P* = 0.05). However, no other differences in vital signs on initial presentation were seen between the 2 groups (Table 2).

SARS-CoV-2 and Other Viral Testing

All patients were tested for SARS-CoV-2 via nasopharyngeal PCR assay. Overall, 20% (21/106) tested positive for SARS-CoV-2 via nasopharyngeal PCR assay (Table 1). Those who had MIS-C were more likely to test PCR positive, compared with the non-MIS-C cohort (OR, 14.48 [4.4–47.73]). Severe acute respiratory syndrome coronavirus 2 antibodies were obtained in 35% (37/106) of the included population. All 11 patients with MIS-C in whom antibodies were obtained were positive compared with only 4 of 26 non-MIS-C patients (100% vs 15.4%, *P* < 0.001). Rapid respiratory panel testing was done on 26 (24.5%) of 106 patients; 4 were positive for viruses. In the MIS-C group, 2 patients were positive for human rhinovirus/enterovirus; in the non-MIS-C group, 1 patient was positive for coronavirus (non-SARS-CoV-2) and another for human metapneumovirus.

Laboratory Investigations

Initial laboratory values for the 2 cohorts are shown in Table 3. Procalcitonin (median [interquartile range], 1.71 [0.22–2.89] ng/mL vs 0.14 [0.06–0.25] ng/mL; *P* < 0.001) and CRP (134.45 [7.95–170.25] mg/L vs 12.6 [1.3–52.8] mg/L; *P* < 0.05) were both

Diagnosis Category (n)	Diagnoses (n = 1 unless otherwise specified)
Acute Viral Illness (45)	Viral illness (36), SARS-CoV-2 (6), gastroenteritis (2), tracheopharyngitis
MIS-C (17)	Multisystem inflammatory syndrome in children (MIS-C)
Bacterial Infection (12)	Urinary tract infection (4), appendicitis, bacteremia, campylobacter jejuni, cellulitis, meningitis, otitis media, pneumonia, streptopharyngitis
Inflammatory Disease (5)	Crohn's disease, dermatomyositis, juvenile idiopathic arthritis, Kawasaki disease, myocarditis
Fever NOS (5)	No formal diagnosis made by team
Asthma (4)	Acute asthma exacerbation
Other (18)	Allergic reaction (2), rash (2), acute disseminated encephalomyelitis, cataplexy, central retinal artery occlusion, congenital heart disease, convulsive syncope, drug reaction, dyspepsia, epilepsy, fatigue, heart failure, mitochondrial disorder, polytrauma, sickle cell crisis

more elevated in patients with MIS-C. No difference was seen in absolute neutrophil count, absolute lymphocyte count, or erythrocyte sedimentation rate. Both aspartate aminotransferase (64 [33.5–82] U/L vs 33.5 [25–47] U/L; $P < 0.05$) and alanine aminotransferase (38.5 [24.5–65] U/L vs 19 [13–27] U/L; $P < 0.05$) had greater elevations in the MIS-C compared with non-MIS-C group. Interleukin 6 levels were obtained in 7 patients (6 MIS-C and 1 non-MIS-C). The median interleukin 6 in the MIS-C group was 77.6 (22.2–142) pg/mL compared with the patient without MIS-C (1.7 pg/mL).

Odds Ratios and ROC Curves for Laboratory Values

Laboratory markers of inflammation were common; elevated procalcitonin in 47 (77.4%) of 66 patients and elevated CRP in 65 (61.3%) of 106 patients were observed (Table 2). An elevated procalcitonin was associated with a higher odds of MIS-C (100%

vs 65.5%; $P < 0.05$). In contrast, MIS-C patients were not more likely than non-MIS-C patients to have an elevated CRP (>8.0 mg/L) on presentation (OR, 2.3 [0.70–7.66]; $P = 0.16$). Cardiac biomarkers were more likely to be elevated in the MIS-C group in both hsTropT (OR, 13 [4.0–42.2]) and N-terminal pro B-type natriuretic peptide (NT-proBNP) (OR, 8.4 [2.3–30.1]). Hematologic laboratory studies were also more likely to be abnormal in the MIS-C cohort including D-dimer (OR, 13 [1.6–103]) and ferritin (OR, 7.8 [2.21–27.33]). Lactate dehydrogenase was also more likely to be abnormal (100% vs 73.2%; $P < 0.05$). Alanine aminotransferase (OR, 3.7 [1.04–13.0]), but not aspartate aminotransferase, was more likely to be abnormal in the MIS-C cohort.

Three laboratory investigations had optimal cutoffs values determined by the Youden index within the normal range for the assay: ferritin (cutoff, 122 μ g/L; upper limit of normal [ULN], 300), NT-proBNP (cutoff, 212 pg/mL; ULN, 450), and alanine aminotransferase (cutoff, 25 U/L; ULN, 55 U/L). In contrast, the cutoff

TABLE 1. Demographics of PUIs for MIS-C

Characteristic	Overall (n = 106)	MIS-C (n = 17)	Non-MIS-C (n = 89)	P
Male, n (%)	59 (55.7)	12 (70.6)	47 (52.8)	0.176
Age, n (%)				0.976
0–5 y	51 (58.1)	9 (52.9)	42 (47.2)	
6–12 y	20 (18.9)	3 (17.6)	17 (19.1)	
13–18 y	20 (18.9)	3 (17.6)	17 (19.1)	
19–21 y	15 (14.2)	2 (11.8)	13 (14.6)	
Weight, median (IQR), kg	21 (11.2–51)	20.8 (11.8–51.4)	21.6 (9.9–50)	0.567
Obese, n (%)	38 (36.9)	8 (47.1)	30 (34.9)	0.342
Previously healthy, n (%)	58 (54.7)	13 (76.5)	45 (50.6)	0.049*
Level of care n (%)				
Ward	71/106 (65.1)	16 (94.1)	55 (61.8)	0.009**
ICU admission	20/71 (28.2)	9/16 (56.3)	11/55 (20)	0.009**
Symptoms at presentation, n (%)				
COVID-19 exposure	16 (15.1)	9 (52.9)	7 (7.9)	<0.001 ***
Cardiovascular (any)	13 (12.3)	1 (5.9)	12 (13.5)	0.688
Constitutional (any)	76 (71.7)	17 (100)	59 (66.3)	0.003**
Dermatologic (any)	38 (35.8)	9 (52.9)	29 (32.6)	0.109
Gastrointestinal (any)	63 (59.4)	14 (82.4)	49 (55.1)	0.036*
Mucocutaneous (any)	8 (7.5)	1 (5.9)	7 (7.9)	0.999
Musculoskeletal (any)	11 (10.4)	1 (5.9)	10 (11.2)	0.999
Neurologic (any)	19 (17.9)	1 (5.9)	18 (20.2)	0.298
Respiratory (any)	47 (44.3)	8 (47.1)	39 (43.8)	0.805
Lymphadenopathy	4 (3.8)	0	4 (4.5)	0.999
Vital signs on emergency department presentation, n (%)				
Tachycardia for age	63 (59.4)	12 (70.6)	51 (57.3)	0.307
Fever present ($>38^{\circ}\text{C}$)	40 (37.7)	10 (58.8)	30 (33.7)	0.050
Hypoxemia ($<90\%$)	8 (7.5)	3 (17.6)	5 (5.6)	0.115
SARS-CoV-2 testing, n (%)				
SARS-CoV-2 nasopharyngeal PCR positive	21 (19.8)	11 (64.7)	10 (11.2)	<0.001 ***
SARS-CoV-2 Ab detected	15 (40.5)	11 (100)	4 (15.4)	<0.001 ***
SARS-CoV-2 PCR or Ab positive	28 (26.4)	17 (100)	11 (12.4)	<0.001 ***
SARS-CoV-2 PCR and Ab positive	8 (7.6)	5 (29.4)	3 (3.4)	0.003**
SARS-CoV-2 PCR positive, Ab not tested	13 (12.3)	6 (35.3)	7 (7.9)	0.002**
SARS-CoV-2 PCR negative, Ab positive	7 (6.6)	6 (35.3)	1 (1.1)	<0.001 ***
SARS-CoV-2 PCR and Ab negative	22 (20.75)	0	22 (24.7)	0.02*

Statistically significant with * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.

ICU indicates intensive care unit; IQR, interquartile range; Ab, antibodies.

TABLE 2. Odds Ratio for Clinical Features and Laboratory Values

Laboratory Values	Reference Range	Odds Ratio (95% CI)
D-dimer, ng/mL	<500 ng/mL	13.0 (1.63–103.54)**
HsTropT, ng/L	<15 ng/L	13.0 (4.00–42.23)***
NT-proBNP, pg/mL	<450 pg/mL	8.38 (2.33–30.08)**
Ferritin, ng/L	20–300 µg/L	7.78 (2.21–27.33)**
Alanine aminotransferase, U/L	10–55 U/L	3.69 (1.04–13.05)*
Aspartate aminotransferase, U/L	22–44 U/L	2.48 (0.82–7.46)
C-reactive protein, mg/L	<8.0 mg/L	2.31 (0.70–7.66)
Clinical features		
Previously healthy		3.18 (0.96–10.5)*
Hospitalized		9.89 (1.25–78)**
ICU admission		5.14 (1.57–16.88)**
COVID-19 exposure		13.17 (3.87–44.9)***
Gastrointestinal symptoms		3.81 (1.02–14.19)*
Fever on presentation		2.81 (0.97–8.12)
SARS-CoV-2 nasopharyngeal PCR positive		14.48 (4.4–47.73)***
Odds ratio could not be calculated for lactate dehydrogenase and procalcitonin because all children with MIS-C had an elevated value.		
Statistically significant with * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.		
ICU indicates intensive care unit.		

for CRP was >14 times the ULN (cutoff, 119.6 mg/L; ULN, 8.0) and resulted in a specificity at that cutoff of 97% (Table 4).

DISCUSSION

Concern surrounding the reopening of schools and moving forward to a “new normal” amid the pandemic is further magnified by potential pediatric disease burden from COVID-19 including MIS-C. Accurate diagnosis of MIS-C from alternative viral and bacterial pathogens is challenging at best based on the available data describing MIS-C presentation variability. Because

many of these laboratory studies, especially ferritin, D-dimer, hsTropT, and NT-proBNP, are not commonly obtained in diagnosing other childhood illnesses (for example, viral gastroenteritis), the question of whether these laboratory abnormalities are unique to MIS-C and can be used to differentiate MIS-C from non-MIS-C illnesses was raised. To our knowledge, this study is the first analysis of the broader population of PUIs for MIS-C focusing exclusively on the emergency department setting. We examined the presenting signs, symptoms, and laboratory values of PUIs for possible MIS-C. Consistent with the existing literature, we found that MIS-C patients more commonly have a COVID-19 exposure, fever, and positivity in PCR or antibody testing. In examining other signs and symptoms and laboratory values, gastrointestinal symptoms, a benign medical history and laboratory elevation of inflammatory, hematologic and cardiac biomarkers may be helpful. Our analysis confirms that patients diagnosed with MIS-C are both more likely to have abnormalities and have higher absolute elevations across numerous laboratory studies, although by cutoff values, only a high CRP appeared helpful in our cohort to distinguish MIS-C from non-MIS-C.

As we learn more about MIS-C and its variable presentation, it may be challenging to identify patients with a progressive clinical course requiring admission or intensive care unit level of support. A provider must feel relatively confident that a patient does not have MIS-C to elect to not admit for further monitoring. Our work only affirms the heterogeneity of the disease, and the ROC analysis indicates that a single laboratory abnormality may not be sufficient in distinguishing MIS-C from other entities. Furthermore, no single laboratory value in our analysis had both a sensitivity and specificity above 90%. It is quite possible that a combination of these signs, symptoms, and laboratory studies may form a diagnostic schema helping further differentiate MIS-C from other diseases.

Presenting clinical signs and symptoms were also shown to be of less value in our analysis. Although more MIS-C patients were febrile and had gastrointestinal symptoms, constitutional and gastrointestinal symptoms were also seen in over half of the patients in the non-MIS-C group. However, gastrointestinal symptoms in this population do appear to differ based on acuity. Carlin et al¹⁷ demonstrated that such symptoms were far less common in outpatients with febrile illnesses than patients referred to the ED with febrile illnesses and MIS-C. Reassuringly, comparing

TABLE 3. Median Laboratory Values on PUIs Either on Presentation or Within 24 Hours of Admission (Median [Interquartile Range])

Laboratory Values	n (%)	MIS-C	n (%)	Non-MIS-C	P
White blood cell count, K/µL	17 (100)	9.8 (5.07–16.82)	86 (96.6)	9.19 (6.64–12.17)	0.763
Absolute neutrophil count, K/µL		3.95 (1.95–12.75)		4.84 (2.45–7.84)	0.826
Absolute lymphocyte count, K/µL		2.37 (0.82–6.67)		2.33 (1.44–4.82)	0.950
Aspartate aminotransferase, U/L	16 (94.1)	64 (33.5–82)	82 (92.1)	33.5 (25–47)	0.008**
Alanine aminotransferase, U/L		38.5 (24.5–65)		19 (13–27)	0.001**
C-reactive protein, mg/L	17 (100)	134.45 (7.95–170.25)	89 (100)	12.6 (1.3–52.8)	0.002**
Erythrocyte sedimentation rate, mm/h	13 (76.5)	36 (11–60)	50 (56.2)	22 (5–36)	0.085
Ferritin, µg/L	16 (94.1)	239 (153–785.5)	77 (86.5)	86 (50–124)	<0.001***
Lactate dehydrogenase, U/L	13 (76.5)	365 (293–386)	71 (79.8)	282 (205–366)	0.035*
Procalcitonin, ng/mL	11 (64.7)	1.71 (0.22–2.89)	55 (61.8)	0.14 (0.06–0.25)	<0.001***
Fibrinogen, mg/dL	15 (88.2)	453 (333–814)	56 (62.9)	394 (300–474)	0.141
HsTropT, ng/L	17 (100)	25 (<6–84)	89 (100)	<6 (<6–7)	<0.001***
NT-proBNP, pg/mL	15 (88.2)	390 (216–1619)	74 (83.1)	123.5 (37–230)	<0.001***
D-dimer, ng/mL	15 (88.2)	1283 (947–3281)	81 (91)	552 (326–1000)	<0.001***
Interleukin-6, pg/mL	6 (35.3)	77.6 (22.2–142)	1 (1.1)	1.7	0.286

Statistically significant with * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.

TABLE 4. Area Under the Curve and Optimal Cutoff Values for Individual Laboratory Values

	Cutoff	Sensitivity	Specificity	AUC (95% CI)
Ferritin, ng/L	122	93.8	72.7	0.85 (0.77–0.93)***
Procalcitonin, ng/mL	0.215	81.8	72.73	0.84 (0.72–0.97)***
NT-proBNP, pg/mL	212	80	74.32	0.80 (0.68–0.93)***
D-dimer, ng/mL	924	80	74.1	0.79 (0.68–0.91)***
Alanine aminotransferase, U/L	25	75	74.4	0.75 (0.64–0.87)**
C-reactive protein, mg/L	119.6	56.3	97	0.74 (0.58–0.9)**
HsTropT, ng/L	16	64.7	88.5	0.73 (0.58–0.88)**
Aspartate aminotransferase, U/L	57	62.5	85.4	0.71 (0.57–0.85)**

Statistically significant with * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.

AUC indicates area under the curve.

their study's cohort, our findings confirm that MIS-C symptomatology seems to differ little between the outpatient and acute care settings.¹⁷

Building upon this work with larger cohorts is needed to further elucidate cutoffs in laboratory values for MIS-C. Others have suggested that NT-proBNP and CRP are more likely to be elevated in MIS-C compared with other febrile childhood illnesses.¹⁷ It is suggested in our analysis that subtle elevations in studies including ferritin and NT-proBNP may not flag as abnormal but may be clinically relevant. Although our study is consistent with prior work in that CRP was more elevated in the MIS-C cohort, it also highlights that elevated inflammatory markers including CRP and procalcitonin are quite common in many childhood illnesses. Although an abnormal CRP value may not be enough to differentiate MIS-C, it would be prudent to use the degree of elevation in inflammatory markers clinically as MIS-C appears to produce a larger inflammatory response than other diseases by laboratory findings. To highlight, a CRP >14 times the ULN (cutoff, 119.6 mg/L; ULN, 8.0) resulted in a specificity of 97% for our cohort though with a compromise in sensitivity. This is consistent with recent data from the CDC suggesting phenotypes of MIS-C with some portion of MIS-C patients appearing to have a larger elevation in inflammatory markers than other groups.¹³ In contrast to the Columbia cohort,¹⁷ our analysis did not observe any difference in blood counts between other febrile childhood illnesses and MIS-C.

Our results must be interpreted within the context of the several limitations of our study. First, the number of MIS-C cases in our cohort is relatively small, consistent with it being a rare diagnosis during the pandemic. Second, as a retrospective chart review from a single institution during a period in which clinical guidelines were evolving, there was variability among laboratory data obtained for patients. Contributing to this variability may have been provider level of concern for MIS-C on initial evaluation, inability to obtain all testing or needed blood volumes in small children, or evolving guidelines and testing recommendations. How MIS-C was defined and described during the study's time frame may very well have represented the "tip of the iceberg" during the initial period of less understanding of mild MIS-C presentations. Third, our study focused on how elements of the initial history, physical, and laboratory assessment of patients correlated with diagnosis of MIS-C, but it does not provide information on the trends of those data, or on severity or outcomes of the MIS-C cases. Fourth, as noted previously, some of the laboratory data (ferritin, NT-proBNP) are both within the normal reference range for our laboratory and demonstrate a statistically significant difference in their absolute values between MIS-C and non-MIS-C groups, which again highlights the clinical spectrum within MIS-C.

Differentiating MIS-C from other diseases of childhood with similar presentations is of the utmost urgency as the COVID-19 pandemic continues to surge into the winter months and emergency departments are besieged by a range of pediatric febrile illnesses. Reassuringly, key laboratory studies and elements of the history used to assist in the diagnosis of MIS-C show promise in also differentiating MIS-C from non-MIS-C and appear largely consistent with similar cohorts. Analyses across a larger cohort will be required to translate these observed differences into clinically useful diagnostic frameworks. Further investigation remains critical in better understanding the comprehensive clinical spectrum of MIS-C patients.

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